Available online at www.sciencedirect.com

ScienceDirect

Physics Procedia 5 (2010) 9–17

Physics Procedia

www.elsevier.com/locate/procedia

LANE 2010

Elucidation of laser welding phenomena and factors affecting weld penetration and welding defects

Seiji Katayama[a*], Yousuke Kawahito[a], Masami Mizutani[a]

[a]*JWRI (Joining and Welding Research Institute), Osaka University*
11-1 Mihogaoka, Ibaraki, Osaka 567-0047, Japan

Invited Paper

Abstract

The behavior and effect of a plasma plume on the weld penetration are greatly different between CO2 laser welding and YAG, disk or fiber laser welding. The effects of the power and the power density on the weld penetration are elucidated. Spattering leading to the formation of underfilled weld beads is controlled by inclining the laser beam. Porosity is formed from bubbles generated from the tip of the keyhole at low welding speed or from the middle part of the keyhole at high laser power density. Cracking easily occurs in pulsed spot welding of aluminum alloys.

Keywords: laser welding; CO_2 laser; fiber laser; laser welding phenomena; weld penetration

1. Introduction

Laser welding has received great attention as a promising joining technology with high quality, high precision, high performance, high speed, good flexibility and low deformation or distortion, in addition to the recognition of easy and wide applications due to congeniality with a robot, reduced man-power, full automation, systematization, production lines, etc.[1]. Applications of laser welding are increasing. The drawbacks of lasers and laser welding are high costs of laser apparatuses, difficult melting of highly reflective or highly thermal-conductive metals, small gap tolerance, and easy formation of welding defects such as porosity in deeply penetrated weld fusion zones.

Therefore, to understand the mechanism of weld penetration and the phenomena in CO_2, YAG or fiber laser welding, a variety of researches have been performed concerning laser-induced plume behavior, the interaction between a laser beam and its induced plume/plasma, melt flows, keyhole behavior and bubble generation in the molten pool leading to the porosity formation in the weld fusion zones [2-26].

2. Experimental Procedure

Welding with high power CO_2 laser, YAG laser or fiber laser was performed on Type 304 austenitic stainless steel or aluminium alloy plates, and their weld penetration and welding phenomena were compared. The behavior and characteristics of a laser-induced plume and/or plasma, and the interaction between a laser beam and a plume were observed or measured by using high speed video cameras, spectroscopic measurement and probe laser beams.

* Corresponding author. Tel.: +81-6-879-8692.
E-mail address: katayama@jwri.osaka-u.ac.jp.

doi:10.1016/j.phpro.2010.08.024

An example of the observation system for visualization of interaction between the laser beam and the plume during welding is schematically shown in Fig. 1 [2,3]. A fiber laser of 1090 nm wavelength is mainly used as a probe beam during YAG or fiber laser welding.

In remote welding, plume behavior, refractive index distribution over the specimen and the effect of a plume on weld penetration were investigated by high speed video cameras and Michelson interferometer, as shown in Fig. 2.

Keyhole behavior, bubble formation, melt flows and molten pool geometry were also observed through X-ray transmission real-time imaging apparatus [4,5]. The observation was performed at the framing rate of 1,000 F/s.

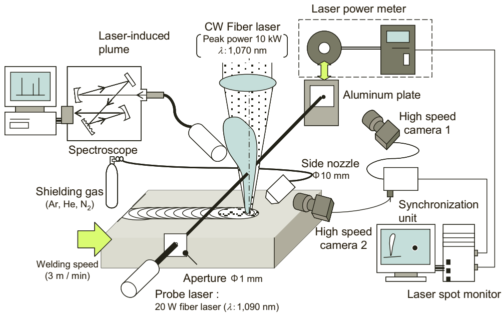


Fig. 1. Schematic of spectroscope, and observation and measurement system for visualization of interaction between probe laser beam and laser-induced plume during laser welding

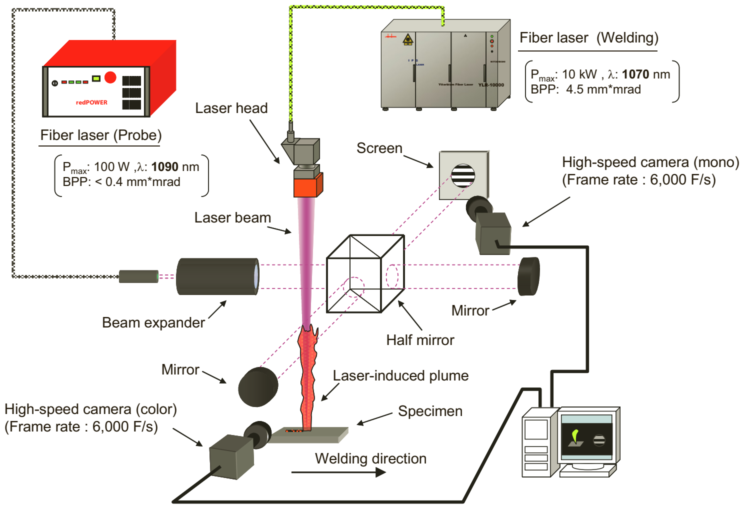


Fig. 2. Schematic arrangement of Michelson interferometer for measuring refraction index.

3. Factors Affecting Laser Weld Bead Geometry and Penetration

In welding with a continuous wave laser, weld penetration and geometry are affected by the kind of lasers with different wavelengths, laser power, beam diameter, beam quality, focused power density, TEM (Transverse Electromagnetic) mode, defocused distance (laser focusing situation), welding speed, materials physical properties, such as laser beam reflectivity, thermal diffusivity, surface tension and the content of volatile elements, environment such as air, shielding gas kind, gas flow rate and vacuum, laser-induced plasma and plume, and so on.[2-26]

Type 304 welds made with a CO_2 laser of 10 kW power in different shielding gases are exhibited in Fig. 3 [4,5]. It shows the effect of a shielding gas on the weld penetration. The penetration is deep in He gas, but the depth decreases with an increase in the gas ratio of Ar to He. The penetration is considerably shallower in N_2 gas than in He gas.

The penetration geometries and depths of weld beads made with a fiber laser of 1 to 10 kW power in Ar shielding gas are shown in Fig. 4 [6]. The penetration increases with an increase in the laser power. Deeply penetrated welds can be produced in fiber laser welding at high powers even in Ar gas. From the comparison of the above figures, the effect of a shielding gas is extremely noted at higher powers of CO_2 laser of about 10.6 μm wavelength, but is not so great in welding with fiber laser of about 1.07 μm wavelength. Thus the development of fiber or disk laser of high powers is expected.

The penetration depths of weld beads produced with fiber laser of 10 kW power and different beam diameters (and correspondingly different power densities) at the focal point are shown as a function of welding speed in Fig. 5 [7]. It is noted that the higher power densities of smaller beam diameters can produce deeper weld beads at high welding speeds. The weld beads are deeper at the slower welding speeds. At low speeds, the effect of power density on weld penetration is small although the power exerts a great effect on the penetration [7,8]. The effect of the defocused distance on the penetration depth of a laser weld bead has been investigated. The weld beads are generally deepest under the condition of the focal point below the specimen surface, but become shallow at the longer defocused distance. It is also understood that deeper penetration can be easily formed in aluminium alloys with a larger amount of volatile alloying elements such as Mg and Zn [9-11]. The deeper penetration can be obtained at the optimum gas flow rate. In vacuum or reduced vacuum, the deepest penetration can be achieved.

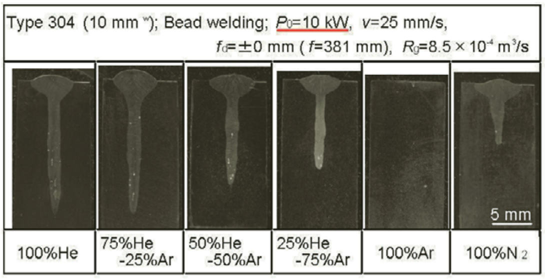


Fig. 3. Cross sections of Type 304 weld beads produced with 10 kW CO_2 laser in different shielding gases

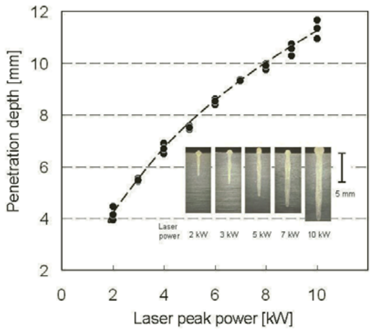


Fig. 4. Penetration geometries and depths of Type 304 weld beads made with fiber laser of 130 μm spot diameter at 1 to 10 kW power and 50

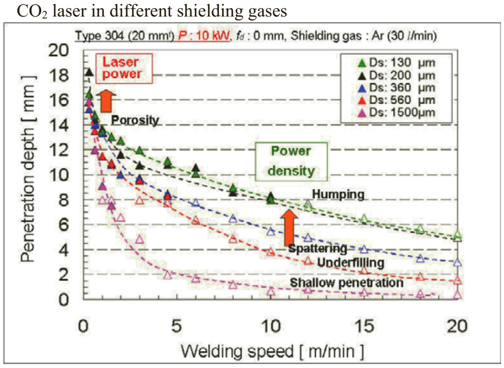


Fig. 5. Effects of beam diameter and welding speed on penetration depths of weld beads produced in Type 304 steel with fiber laser of 10 kW power

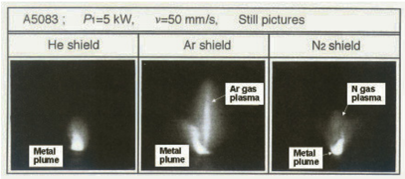


Fig. 6. Plasma and plume observed during CO_2 laser welding of A5083 alloy in He, Ar and N_2 coaxial shielding gas, showing effect of shielding gas kind on plasma formation

4. Behaviour and Characteristics of Laser-Induced Plume, and Interaction between Laser Beam and Induced Plume during CW Laser Welding

Examples of laser-induced gas plasma and metal plume observed during CO_2 laser welding of A5083 aluminium alloy under the conditions of 5 kW power, 50 mm/s speed and 30 l/min coaxial flow of He, Ar and N_2 shielding gas are shown in Fig. 6 [12]. In He gas a plume is only formed by the emission of neutral metal atoms spouting from a keyhole inlet during welding, while in Ar or N_2 gas, gas plasma is always or sometimes formed under the nozzle in addition to the plume from the keyhole, respectively. In welding with CO_2 laser of 10 kW or more in Ar or N_2 side gas, a gas plasma flows up to the laser beam periodically to block an incident laser beam so as to reduce weld penetration [13]. It is interpreted that Ar or N_2 gas plasma is more easily formed because of lower ionisation potentials in comparison with He gas, and the gas plasma exerts the inverse bremsstrahlung process on the CO_2 laser beam because the effect is proportional to λ^2, where λ is the laser wavelength [14,15].

An example of spectroscopic measurement and analysis of a plume during welding of Type 304 stainless steel with a 10 kW fiber laser beam at the ultra-high power density of about 1 MW/mm^2 in Ar shielding gas is shown in Fig. 7 [6]. Almost all peaks come from the emission of neutral atoms, and the emission from Ar gas is not detected. The inverse bremsstrahlung effect seems to be negligibly small in welding with a laser beam of about 1 μm wavelength. According to Saha equation, it is estimated that a plume is in the state of a weakly-ionized plasma (about 6000 K and 2% atomic ions) [6]. Figure 8 shows examples of photos displaying the relationship between plume behavior and probe laser motion [2, 6]. It was confirmed that the refraction and attenuation of a probe laser beam existed due to an inclined refraction index profile caused by slanted distribution of vapors density at high temperatures and Rayleigh scattering, respectively [2, 11, 16, 17]. The interaction of such a small-sized plume on a laser beam of about 1.06 μm wavelength was judged to be small.

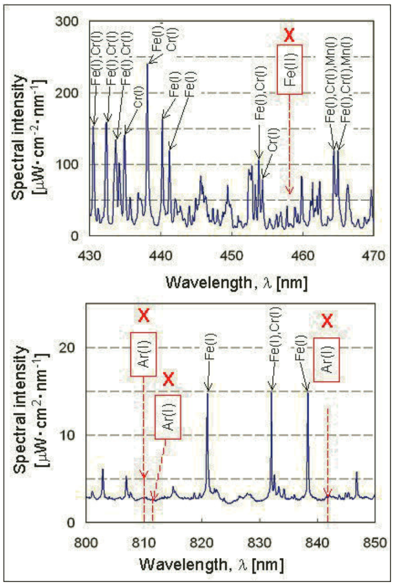


Fig. 7. Spectroscopic measurement and analytical result of plume induced during 10 kW fiber laser welding of Type 304 stainless steel.

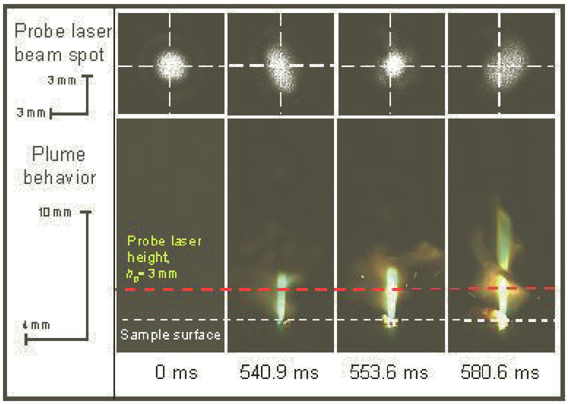


Fig. 8. Simultaneous observation results of probe fiber laser behavior (upper) and laser-induced plume (lower), showing visual interaction between laser beam and laser-induced plume.

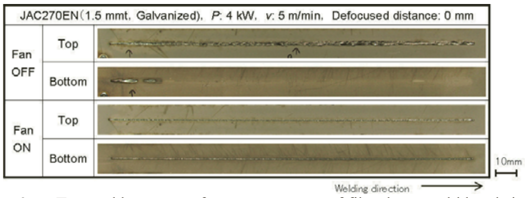


Fig. 9. Top and bottom surface appearances of fiber laser weld beads in zinc coated steel sheets at 4kW and 5 m/min with/without fan blowing (Initial bead spot size: about 0.4 mm).

Experimental results obtained during fiber laser remote welding of 270 MPa steel under the conditions of 4 kW and 5 m/min are shown in Fig. 9. The welds were produced with and without fan blowing. The upper photographs show the change in welding mode from full penetration to partial penetration at about 25 mm from the welding start point in zinc-coated steel. However, the lower photographs show that a full penetration weld forms all over the weld length. Therefore, the plume behavior was observed through high speed video. Figure 10 shows examples of observation pictures of laser-induced plume and molten pool in the elapse time of 1.1 s in the case of air and fan blowing. Without fan blowing, a laser-induced plume glows up to over 100 mm and melted areas are formed in front of or at the sides of the laser irradiation spot. With fan blowing, in contrast, the plume is low and a keyhole is formed in a small molten pool. It is considered that a highly grown plume and the consequent low refractive index zone exert a great influence on the welding results. A tall plume over the specimen is heated, and thus the distribution of the refractive-index is so different that the incident laser beam is defocused, deflected, refracted and/or inclined. As a result, the transition (the decrease in the focused condition) in weld penetration depth occurred in air without fan-blowing. Blowing with a fan was beneficial. It is concluded that the effect of a small plume is small but that of a tall plume is great during fiber laser welding.

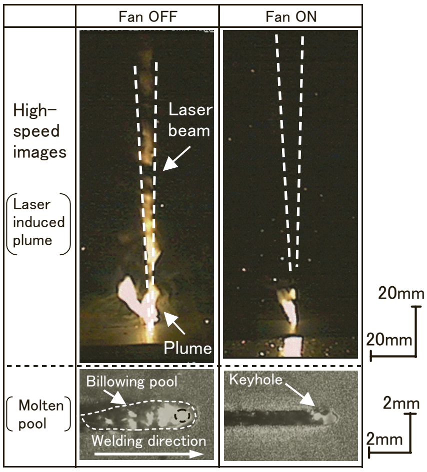


Fig. 10. Laser-induced plume and molten pool observed during laser remote welding of steel with and without fan

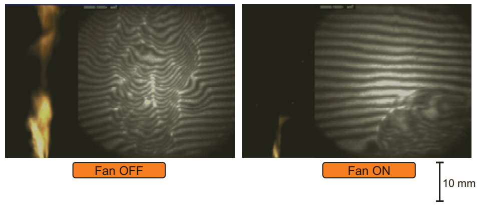


Fig. 11. Plume behavior and refractive index distribution during remote laser welding of steel without and with fan blowing

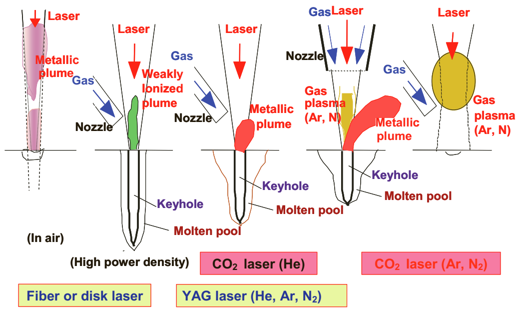


Fig. 12. Schematic illustration of plume and plasma behaviour and consequent keyhole and/or melting in laser welding.

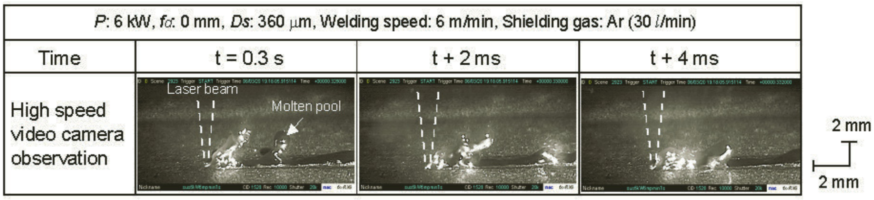
The plume and plasma behavior and the consequent keyhole geometry in the molten pool are schematically illustrated in Fig. 12 [15]. In the case of CO_2 laser of about 10.6 μm wavelength, Ar gas and N_2 gas forms gas plasma, resulting in the apparent decrease in weld penetration, whilst in the case of YAG, fiber or disk laser of about 1 μm wavelength, the shielding gas effect is small, and laser power and its density exert the penetration [10, 16]. However, the effect of a tall plume should be noted and be reduced in remote laser welding.

5. Formation Mechanisms and Suppression Procedures of Laser Welding Defects

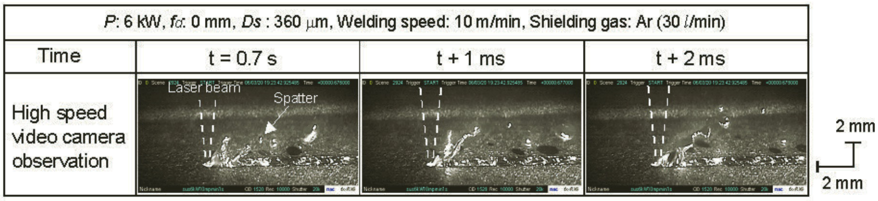
Welding can be carried out under a variety of conditions by using a high-brightness and high-quality fiber laser. In welding with a moderate high power density of fiber laser, sound weld beads with convex surfaces were produced. On the other hand, underfilled weld beads were formed due to the generation of large spatters at the high welding speed of more than 10 m/min at the lower power density, while the humping weld beads were produced at high power density (See Fig. 5 as a reference). To clarify these welding phenomena, high-speed video observation of the surface of a molten pool was performed.

Fiber laser bead-on-plate welding was performed under the welding conditions of 6 kW laser power, ϕ 360 μm spot diameter and various welding speeds. Examples of the observation results at the welding speed of 6 m/min and 10 m/min are shown in Fig. 13 (a) and (b), respectively. At 6 m/min, the molten pool was more than 7 mm long. A keyhole was located near the front of the molten pool and the melt flowed out from the keyhole inlet to form a convex surface. Large spatters flowed away from the melt, but jumped on the large molten pool and disappeared in the pool. At the welding speed of 10 m/min, the molten pool became smaller and the spattering became more severe. Almost all melts near the top surface formed spatters. Consequently, an underfilled weld bead with large spatters was formed. This means that spattering can be prevented by including jumped melts in a broad molten pool.

Figure 14 shows the observation results of the molten pool under the welding conditions of 6 kW laser power, ϕ 115 μm spot diameter and 6 m/min welding speed. A Xenon arc lamp was used for illumination to observe the molten pool and laser-induced plume. It was confirmed that the molten pool was very narrow and long, the backward melt flow due to the ejection of laser-induced plume was strong and the humping part was generated gradually at the rear part of molten pool. It was considered that humping was formed due to strong backward melt flows induced by laser-induced plume and high surface tension with a very narrow bead width.



(a) At welding speed of 6 m/min



(b) At welding speed of 10 m/min

Fig. 13. High-speed video camera observation results of molten pool surface during fiber laser welding at 6 kW power and 360 μm beam diameter

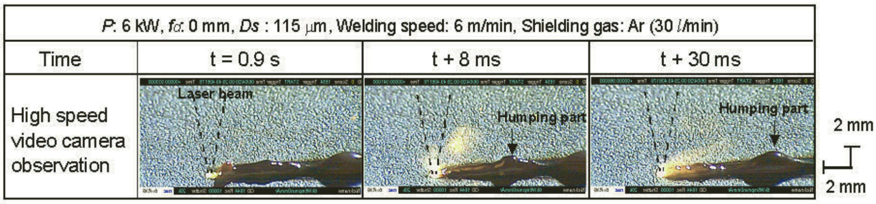


Fig. 14. High-speed video camera observation results of molten pool surface at high power density

Porosity is easily formed in deeply penetrated weld beads at low welding speeds. X-ray transmission observation results confirm that bubbles are generated from a keyhole tip to form porosity at low welding speeds while bubbles are not formed at high welding speeds, resulting in no porosity, as examples of 6 kW fiber laser welding are shown in Fig. 15 [11]. From such observation results, plume behavior, keyhole behavior, melt flows, and bubble generation leading to porosity formation in the molten pool are schematically summarized in Fig. 16 [15]. At low welding speeds, a deep keyhole is likely to collapse, then a laser beam is irradiated on the collapsed liquid wall of the keyhole, and consequently downward melt flows are caused by the recoil pressure of evaporation from the collapsed wall. Such phenomena may produce large bubbles leading to pores. As the welding speed is slightly increased, intense evaporation sometimes takes place from the front bottom wall of a keyhole (where the liquid layer is thin) to deform the rear wall of the keyhole (where the liquid is large) so as to produce a bubble. Under a certain condition, bubbles flow upward to disappear from the molten pool surface. If the material has such a high surface tension to produce a stable keyhole, the vapors are strongly ejected upwards from the keyhole inlet to induce the melt flow near the inlet upward, and smaller bubbles are formed from the keyhole tip. Consequently pores are present near the bottom part of the weld bead. When the welding speed is high enough, a plume ejects upwards, and a keyhole becomes so stable as to suppress or prevent bubble formation, resulting in reduced or no porosity. At high powers such as 20 to 40 kW CO_2 laser in He shielding gas and/or at high power densities such as 10 kW fiber laser of 0.2 mm focused beam, bubbles leading to pores are sometimes generated from the middle part of a keyhole.

Examples of X-ray transmission photos during cw YAG laser welding of Type 304 steel in Ar, He and N_2 shielding gas are shown in Fig. 17 [18]. Bubbles are formed in an inert gas such as Ar and He, but bubbles are hardly seen in N_2. Ar and He gas is generally detected in the pores. Therefore, Ar or He gas is involved in an unstable keyhole at lower welding speeds, but a harmful influence of N_2 is reduced by reacting with Cr vapors in the keyhole or being included due to high solubility in the molten pool. N_2 gas is regarded as a proper shielding gas in YAG, disk or fiber laser welding of austenitic stainless steels.

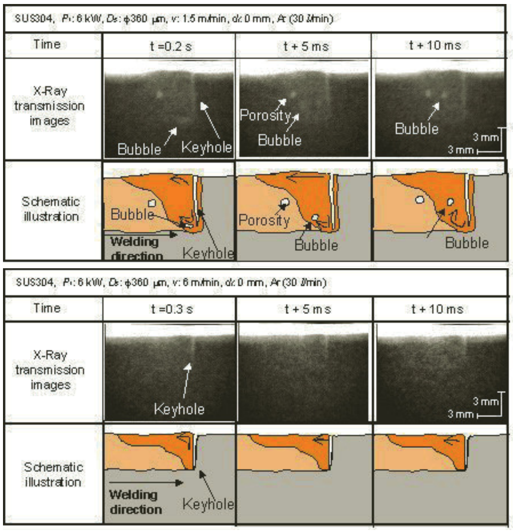


Fig. 15. X-ray transmission observation results and schematic representation of keyhole behaviour, melt flows, bubble formation and molten pool geometry during welding with 6 kW fiber laser at speeds of 1.5 and 6 m/min

Bubbles, resulting in pores or porosity, are normally formed from the tip of a keyhole during laser welding, and thus the following procedures for the reduction in porosity are proposed [9, 13, 19-21], and their effectiveness is confirmed: (1) Full penetration welding, (2) proper pulse modulation [22, 23], (3) forwardly tilted laser beam, (4) twin-spots laser welding, (5) vacuum welding, (6) the use of tornado nozzle, (7) hybrid welding with laser and TIG or MIG at high arc current, (8) high speed welding.

In laser full penetration welding of 12 mm thick steel plate, porosity and underfilling is present at high laser power density, whilst such porosity can be reduced or prevented by utilizing a focusing optic with a moderate power density, as compared in Fig. 18. In laser welding of thick steel plate with some gap, underfilling or humps on the bottom surface can also be easily formed. Such defects could be prevented by using a hot wire instead of a cold wire [24].

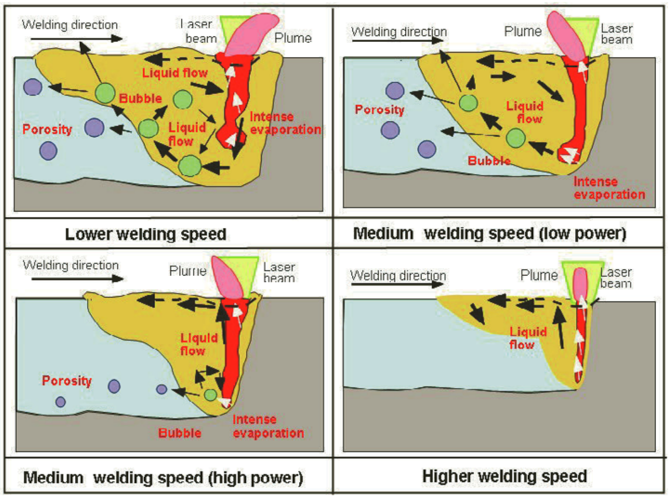


Fig. 16. Schematic representation of plume behavior, keyhole behavior, melt flows, and bubble generation leading to porosity formation in molten pool according to respective conditions

In laser welding of Zn-coated steel, it is important to control a gap between lapped sheets in order to escape vapors and their pressure due to low evaporation temperature and high vaporization pressure of Zn. Laser brazing can be applied to join Zn-coated steel sheets without porosity or pits [25]. Moreover, laser welding of die cast alloys or sintering alloys is difficult because pores are inevitably formed in weld fusion zones [25].

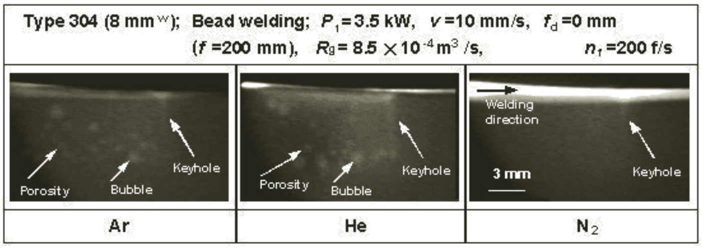
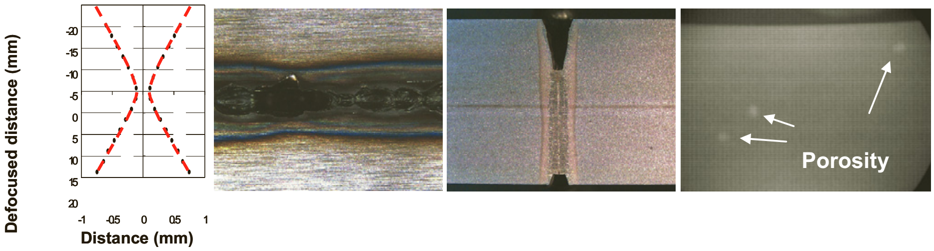
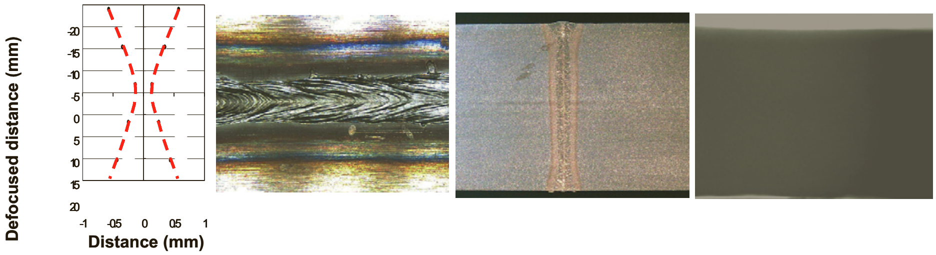


Fig. 17. X-ray transmission observation photos of keyhole and/or bubble generation in molten pool during cw YAG laser welding of Type 304 steel in Ar, He and N_2 coaxial shielding gas



(a) Focusing situation (b) Surface (c) Cross section (d) X-ray inspection result



(e) Focusing situation (f) Surface (g) Cross section (h) X-ray inspection result

In some aluminum alloys, for example, cracks occur under some conditions, but do not take place under the other conditions. It is found that cracking is easily formed at higher welding speeds in thicker plates or in pulsed spot welding. In laser welding of aluminum alloys, spattering occurs easily, and consequently an underfilled weld bead can be formed. The surface is modified by applying hybrid welding with MIG arc (as the following heat source) [22,26].

6. Conclusions

Laser welding phenomena are satisfactorily understood. In this paper, welding results made with continuous wave high power lasers are described. In the case of welding with CO_2 laser of 10.6 μm wavelength, Ar or N gas plasma are easily formed to absorb laser irradiation energy, while in the case of welding with YAG, disk or fiber laser, a tall plume and its upper part of low refraction index should exert an effect of refraction, defocusing and Rayleigh scattering to reduce penetration. Spattering and humping phenomena are observed to occur due to the effect of laser-induced plume. Porosity is chiefly generated from the tip of a deep keyhole, and some preventive procedures are developed. Cracking occurs in laser weld metals of thicker aluminum alloy plates at high welding speeds or in pulsed spot welding of aluminum alloys.

Acknowledgements

We would like to acknowledge Dr. Naoki Seto, Mr. Keisuke Kinoshita, Mr. Naoki Matsumoto, Mr. Shimpei Oiwa and so on of former Graduate Students of Osaka University.

References

1. S. Katayama, 'New development in laser welding', in 'New developments in advanced welding', ed. by Ahmed N, 2005, Cambridge England, Woodhead Publishing Limited, 158-197.
2. Y. Kawahito, K. Kinoshita, S. Katayama, S. Tsubota and T. Ishide, 'Visualization of Interaction between Laser Beam and YAG-Laser-Induced Plume', ICALEO 2005 Congress Proceedings, Miami, LIA (Laser Institute of America), 98 (2005) 920-928 (CD).
3. Y. Kawahito, M. Mizutani and S. Katayama, 'Elucidation of high-power fiber laser welding phenomena of stainless steel and effect of factors on weld geometry', J. Phys. D: Appl. Phys., 40 (2007) 5854-5859.
4. S. Katayama, N. Seto, J.D. Kim and A. Matsunawa, 'Effect of nitrogen on laser weldability of aluminium alloys', IIW, (1999), IIW Doc No. IX-1949-99.
5. N. Seto, S. Katayama and A. Matsunawa, 'High-speed simultaneous observation of plasma and keyhole behavior during high power CO_2 laser welding: Effect of shielding gas on porosity formation', J. of Laser Applications, 12-6 (2000), 245-250.
6. Y. Kawahito, K. Kinoshita, N. Matsumoto, M. Mizutani and S. Katayama, 'Interaction between Laser Beam and Plasma/Plume Induced in Welding of Stainless Steel with Ultra-High Power Density Fiber Laser', Q. J. Jpn. Weld. Soc., 25-3 (2007) 461-467. (in Japanese)
7. S. Katayama, Y. Kawahito, K. Kinoshita, N. Matsumoto and M. Mizutani, 'Weld Penetration and Phenomena in 10 kW Fiber Laser Welding', ICALEO 2007 Congress Proceedings (Proc. of 26th International Congress on Applications of Lasers & Electro-Optics), Orlando, LIA, 100 (2007) 353-359. (CD)
8. K. Kinoshita, M. Mizutani, Y. Kawahito and S. Katayama, 'Phenomena of Welding with High-Power Fiber Laser', ICALEO 2006 Congress Proceedings, (25th International Congress on Applications of Lasers & Electro-Optics), Scottsdale, LIA 99 (2006) Paper #902 535-542. (CD)
9. S. Katayama and C.D. Lundin, 'Laser welding of aluminum alloy 5456 – Laser weldability of aluminum alloys (Report I) –', J. of Light Metal Welding and Construction, 27-7 (1991) 295-307. (in Japanese)
10. S. Katayama and C.D. Lundin, 'Laser welding of commercial aluminum alloys – Laser weldability of aluminum alloys (Report II) –', J. of Light Metal Welding and Construction, 29-8 (1991) 349-360. (in Japanese)
11. A. Matsunawa, K. Kojima and S. Katayama, 'CO_2 laser weldability of aluminum alloys (Report I) –Effect of welding conditions on melting characteristics–', J. of Light Metal Welding and Construction, 35-10 (1997) 461-472. (in Japanese)
12. Matsunawa A., Seto N., Kim J.D., Mizutani M. and Katayama S., 'Observation of keyhole and molten pool behaviour in high power laser welding –Mechanism of porosity formation and its suppression method-', *Trans. of JWRI*, 30-1 (2001) 13-27.
13. N. Seto, S. Katayama and A. Matsunawa, 'High-speed simultaneous observation of plasma and keyhole behavior during high power CO_2 laser welding: Effect of shielding gas on porosity formation', *J. of Laser Applications*, 12-6 (2000) 245-250.
14. A. Matsunawa, J.D. Kim, S. Katayama and T. Takemoto, 'Spectroscopic studies on laser-induced plume of aluminum alloys', Proc. ICALEO '95, San Diego, LIA, 80 (1995) 719-728.
15. S. Katayama, Y. Kawahito and M. Mizutani, 'Collaboration of physical and Metallurgical viewpoints for understanding and process development of Laser Welding', *ICALEO 2007 Congress Proceedings* (Proc. of 26th International Congress on Applications of Lasers & Electro-Optics), Orlando, LIA, 100 (2007) #701 360-369. (CD)
16. Y. Kawahito, K. Kinoshita, N. Matsumoto, M. Mizutani and S. Katayama, 'High-speed observation and spectroscopic analysis of laser-induced plume in high-power fiber laser welding of stainless steel', Q. J. Jpn. Weld. Soc., 25-3 (2007) 455-460. (in Japanese)
17. S. Katayama, S. Oiwa, H. Ishida, N. Ozawa, M. Mizutani and Y. Kawahito, 'Effect of Laser-Induced Plume on Laser Beam Focusing during Remote Welding', 61st Annual Assembly of Int. Inst. Welding (IIW), Graz, IIW Doc. IV-962-08.
18. S. Katayama, Y. Kawahito and M. Mizutani, 'Effect of gas compositions on laser weldability of stainless steel', Proc. of the 69th Laser Materials Processing Conference, 69 (2007) 161-170. (in Japanese)
19. S. Katayama, Y. Kobayashi, N. Seto, M. Mizutani and A. Matsunawa, 'Effect of Vacuum on Penetration and Defects in Laser Welding', Proc. of Laser Materials Processing Conference ICALEO 2000, LIA, USA, 89 (2000), Section C 182-191.
20. S. Katayama, N. Seto, J.D. Kim and A. Matsunawa, 'Formation Mechanism and Reduction Method of Porosity in Laser Welding of Stainless Steel', Proc. of ICALEO '97, San Diego, USA, 83-Part 2 (1997) Section G 83 - 92.
21. S. Katayama, N. Seto, J.D. Kim and A. Matsunawa, 'Formation Mechanism and Suppression Procedure of Porosity in High Power Laser Welding of Aluminum Alloys', Proc. of ICALEO '98, LIA, Orlando, USA, 85 (1998) Section C 24-33.
22. S. Uchiumi, J.B. Wang, S. Katayama, M. Mizutani, T. Hongu. and K. Fujii, 'Penetration and Welding Phenomena in YAG Laser-MIG Hybrid Welding of Aluminum Alloy', ICALEO 2004 Congress Proceedings, San Francisco, LIA, 97 (2004), P530. (CD)
23. S. Katayama and A. Matsunawa, 'Laser weldability of aluminum alloys', Proc. of the 69th Laser Materials Processing Conference, 43 (1998) 33-52. (in Japanese)
24. T. Onishi, M. Mizutani Y. Kawahito and S. Katayama: 'High-power Fiber Laser Butt Welding of Thick High-strength Steel Plate using Sensing System with Hot Wire', *Proc. of 28th International Congress on Applications of Lasers & Electro-Optics (ICALEO 2009)*, 2009, 773-779. (CD)
25. S. Katayama, Y. Kawahito and M. Mizutani, 'Welding phenomena and results in laser joining of various materials', *Proc. of the 66th Laser Materials Processing Conference*, 66 (2006) 1-8. (in Japanese)
26. H. Nagayama, M. Mizutani, Y. Kawahito and S. Katayama, 'Weld Penetration and Welding Phenomena of Aluminum Alloy with High-Power Fiber Laser', ICALEO 2007 Congress Proceedings (Proc. of 26th International Congress on Applications of Lasers & Electro-Optics), Orlando, LIA, 2007 100 916-924. (CD)